Investigation into how weekly earnings depend on different variables

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Abstract

Many social factors affect a person's weekly wages. Most notably, a person's highest form of education seems to be the driving force behind the disparity with pay. Unfortunately, there also seems to be racial inequality in pay however one must note also this data set had $\times 8$ more white people than black people and hence it might be a skewed measure. Surprisingly, while one might assume age is a seminal factor, it's positive contribution to one's wages are minute. Finally, the location of one's employment significantly impacts their wages; this makes sense as all the high paying jobs aren't expected to be equally distributed across the nation.

1 Introduction

This report will investigate how weekly earnings of individuals in the United States vary based on demographic factors including race, age, education, and region of employment. Using a subset of data from 2011 in a specific region, we attempt to develop and interpret statistical models to understand the primary factors influencing earnings. This will be done by exploring how the different factors individually affect the wages and then by fitting a linear model and assessing the variables and interpreting the effects of the variables.

Summary	
Variable	**Description**
Region	Categorical — Region of employment — Levels: Northeast, Midwest, South, and West — Baseline: Midwest
Location	Categorical — Metropolitan status of an individual's employment — Levels: Metro, and Not Metro — Baseline: Metro
Age	Numerical — Individual's age measured in years
Education Category	Categorical — The highest educational qualification of an individual — Levels: 16 ranging from "Less than First grade" to "Doctorate Degree" — Baseline: AssocDegAcadem
Education Code	Integer — Ranging from 1 to 16, corresponding roughly with increasing level of education
Race	Categorical — Levels: White, and Black — Baseline: Black
Wage	Numerical — A person's weekly earnings measured in dollars

2 Exploratory Data Analysis

We are exploring how each of these variables affects wages. Figures 1-4 are box-plots of wages against the respective categories with the dotted-red line representing the average wage of all individuals which is 1056.4 dollars. Note that all measure of mean, median and IQR are in dollars.

Figure 1 seems to show that the means and medians in decreasing order are Northeast, West, Midwest, and South suggesting a possible geographic disparity in wages. But also note that areas with higher median and mean wages have higher variation in the wages; when looking at the IQR, they are $906.834, 726, 200$ with respect to Northeast, West, Midwest and South.

Figure 2 shows that metropolitan jobs have a higher median and mean wage than non-metropolitan jobs. However, with the higher mean and median wage, there is also a higher IQR (838 for Metro and 582 for Not Metro) which indicates higher variability in wages. With the bottom of the boxes and whiskers being about the same this therefore suggests that for metropolitan jobs there are more higher paying ones. In all, this leads us to believe that location of job does affect wages (for this sample anyway).

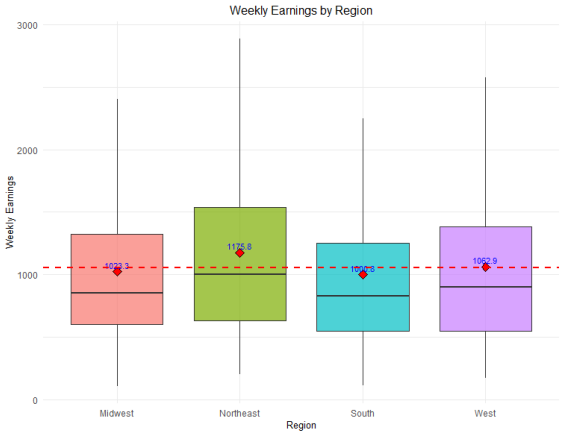


Figure 1: Box-plot of wages against region of employment with mean highlighted for each class

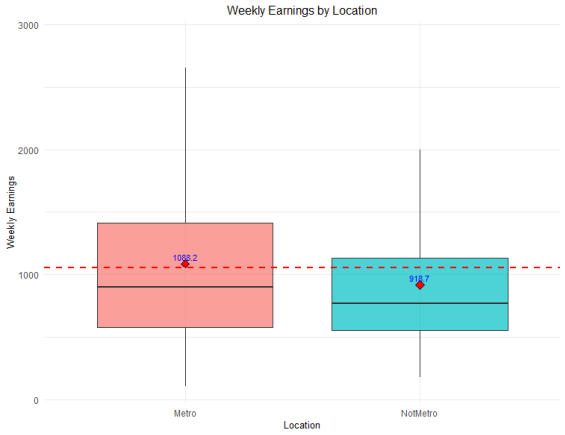


Figure 2: Box-plot of wages against metropolitan status of employment with mean highlighted for each class

Figure 3 shows a major wage disparity due to race; the mean wage for white people is 1077.5, which is higher than the average wage, while the mean wage for black people is 822.3, which is significantly lower than the average wage. This suggests that race perhaps plays a role in the pay of an individual from this sample.

Figure 4 strongly shows that people with higher levels of education (indicated with a larger integer value) earned higher wages. This general increasing trend can be seen as the means in Figure 4 are mostly increasing. Similar to the other variables, for higher average wages there's also higher variability in the wages as indicated by the increasing IQR (for the lower education codes there's IQR ranging from $255 - 558$ whilst for higher educational codes the IQR ranges from $981 - 1635$).



Figure 3: Box-plot of wages against race with mean highlighted for each race

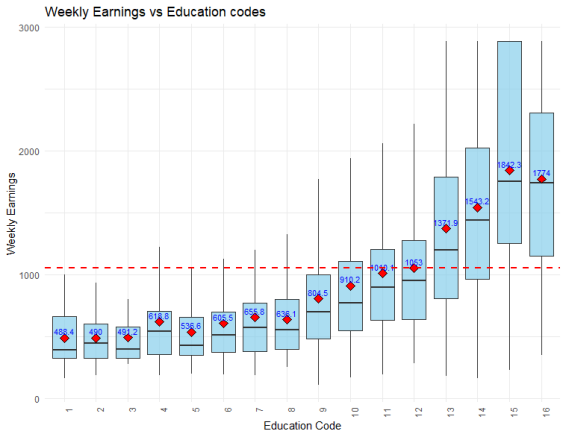


Figure 4: Box-plot of wages against the educational codes where the integers are categories with each code's mean highlighted. The educational codes are an ordered list of the educational categories

Figure 6 shows a general positive whereby older individuals earned more on average than younger individuals. However, after a certain age (mid 60s) the pay for older people seemed to be lower. This is understandable as they are likely retired. However this still shows that age and seniority plays a crucial part

in an individuals weekly wages. Figure 5 is the same as Figure 6 without taking average wages for each, this shows that variability in wages; younger people's wages didn't vary much however older people's wages varied quite considerably simply by looking the spread in Figure 5.After looking at these statistical summaries we can see that there is reason to believe that these explanatory variables affected wages.

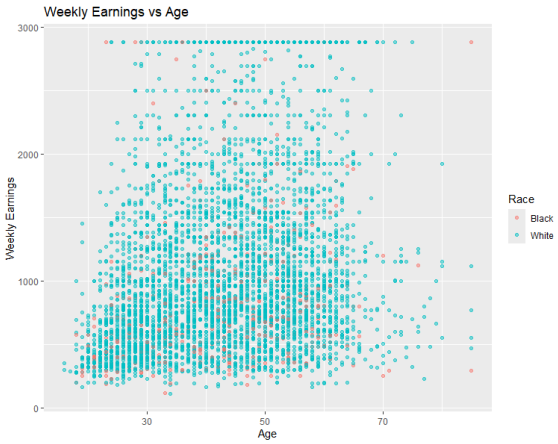


Figure 5: Scatter plot for wages against ages with the races highlighted

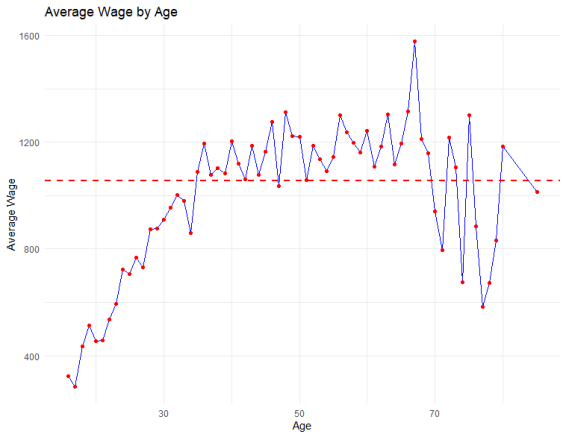


Figure 6: Scatter plot for average wages for each age plotted against ages of the individuals

3 Modelling

We start by fitting a normal linear model which includes the main effects. We assume an additive errors and those errors are independently and distributed with a normal distribution centred at 0 and with a constant variance. This means that the model looks like:

$$\text{Wage} = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{EducCode}) + \beta_3 \mathbb{1}_{\text{Location=NotMetro}} + \beta_4 \mathbb{1}_{\text{Race=White}} + \sum_{i=1}^{3} \gamma_i \mathbb{1}_{\text{Region=element i of } \mathbf{Region}} + \epsilon$$

where **Region** is just the set with elements Northeast, South, Midwest in that order. As stated in the beginning, the baseline factors for Location, Race, and Region are Metro, Black, and Midwest. Here we're using Educational code as a numerical variable to simplify the model and provide interpretable results.

4 Model Diagnostics and Revision

4.1 Diagnostic Checks

Before interpreting and analysing the fitted coefficients, we must check if such a model is valid. Figure 7a (left), has a widening funnel shape which means that the variance of the studentised residuals is increasing as the fitted values increase. This suggests non-linearity which violates the assumption of homoscedasticity (equal variance). Moreover, there seems to be a pattern in some of the studentised residuals where it forms a linear decreasing line; this could indicate multicollinearity (some of the variables are dependent on each other partially or wholly) and suggest that we perhaps need to use a transformation. This is further supported in Figure 7b (right) as the larger values deviate from the normal qq-line, suggesting a violation of the homoscedasticity assumed in the initial model. We therefore require a transformation that can stabilise the variance such as a Box-Cox transformation.

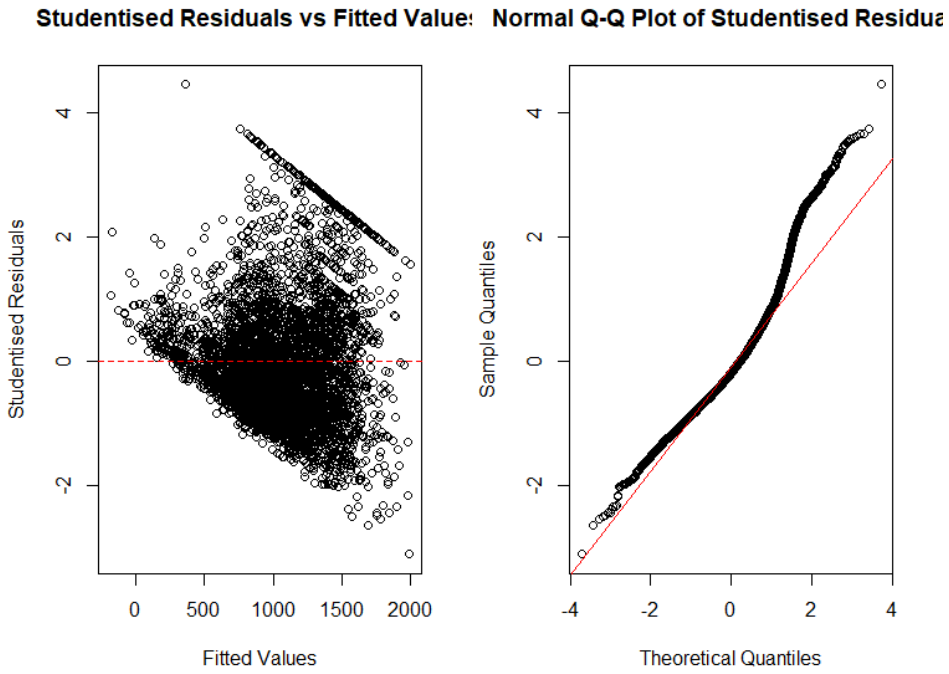


Figure 7: 7a (left) shows the how the studentised residuals vary for all the fitted values 7b (right) explores if the sample quantiles of the studentised residuals follows a normal qq plot

4.2 Model Revision and Selection

The Box-Cox transformation aims to find the λ such that $(\text{wage})^{\lambda}$ fits the the normal linear model. The λ that maximises this likelihood is -0.02 and hence (Figure 8), to have an interpretable model, we will choose $\lambda = 0$ which corresponds to a transformation:

$$\text{wage} \xrightarrow{\lambda=0} \ln(\text{wage})$$

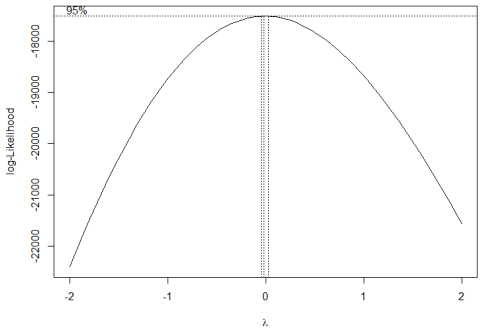


Figure 8: Profile log-likelihood for parameter λ. The λ that maximises the log-likelihood is $\lambda = -0.020202$

Hence the full fitted model after this transformation becomes:

$$\ln(\text{Wage}) = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{EducCode}) + \beta_3 \mathbb{1}_{\text{Location=NotMetro}} + \beta_4 \mathbb{1}_{\text{Race=White}} + \sum_{i=1}^{3} \gamma_i \mathbb{1}_{\text{Region=element i of } \textbf{Region}} + \epsilon$$

This new transformed model yields studentised residuals more evenly spaced and hence it does not violate homoscedasticity as seen in Figure 9a (left). Moreover, the Normal Q-Q plot is closer to the theoretical straight line, once again indicating that there is not a violation of homoscedasticity.

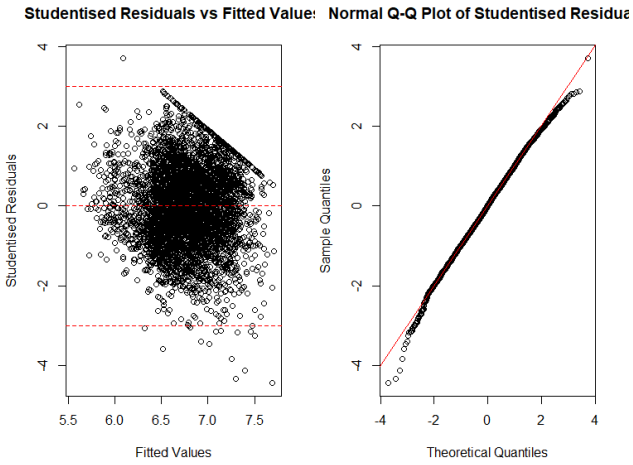


Figure 9: Studentised residuals vs fitted values (9a-left) and normal Q-Q plot of studentised residuals (9b-right) of the transformed wages model

In addition, more of the studentised residuals are within the $(-3, 3)$ range compared to the prior model. It also would be wise to identify influential points and removing them as influential high leverage points can skew the model making the fitted coefficients less reliable. To do this we look at the points that have a Cook's distance such that:

$$C_k \geq \frac{8}{n - 2p}$$

and we remove them.

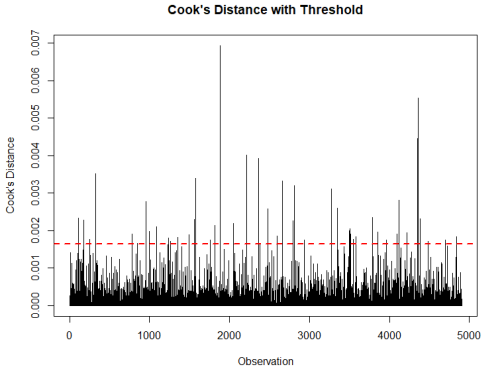


Figure 10: Cook's distances for the revise model. The red line is the threshold and hence the 49 points with a Cook's distance above this red line will be removed.

This results in a cleaned and transformed normal linear model of our original set up. Figure 11 shows that even fewer points have studentised residuals exceed the range of $(-3, 3)$, despite the cleaning process resulting in minimal change, this change resulted in a better model.

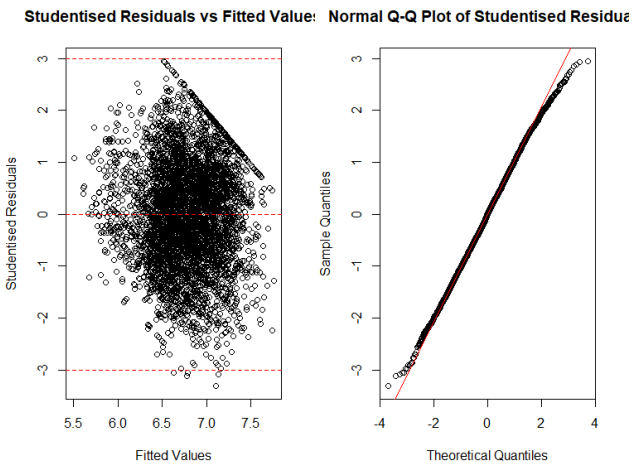


Figure 11: The plots for the studentised residuals for the cleaned and transformed normal linear model.

5 Education Variable: Categorical or Numerical

We want to explore which model is better: model with Education code (numerical) or model with Educational categories (categorical). There are different methods to analyse model choice. Initially, we must note that in a scenario like this, ANOVA with not a good method to ascertain such a choice as ANOVA checks if the average values (means) of a variable differ across predefined groups. It assumes that the groups these groupings already exist. However, we're testing if the groupings exist meaning we're unsure whether the variable should even be treated as groups (categorical), this assumption therefore doesn't make sense. Hence other methods such as AIC (Akaike Information Criterion) and BIC (Bayes Information Criterion). Each one has its own benefits but generally:

- AIC focuses on predictive accuracy with lower penalities. Its use cases are best suited for small datasets or when the actual model may not be in the set
- BIC focuses more on model simplicity with a higher penalty on more complicated models. Its use cases are best suited for large datasets or when true model is believed to be in the given set of models

Clearly here, the use of BIC is more apt as we have a large dataset and the true/desired model is believed to be one of the two being compared. Moreover, this entire analysis focuses on model simplicity and inference as opposed to predicative power of the model.

$$\text{BIC} = -2l(\hat{\beta}, \hat{\sigma^2}) + p \ln(n)$$

where n is the number of data points and p is the number of explanatory variables. Therefore we get:

$$\text{BIC}_{\text{num}} = 7366.192 \quad \text{and} \quad \text{BIC}_{\text{catg}} = 7435.588$$

and hence we should use the numerical model for the Education measure (EducCode) as it minimises the BIC.

6 Interpretation and Conclusions

Finally, using the cleaned and transformed normal linear model to fit the data, we get:

Table 1: Coefficients of the Cleaned and Transformed Normal Linear Model taking a 95 percent significance level and their interpretations

Variable	Coefficient	Std. Error	P-Value	Interpretation
Intercept	4.910	0.046	< 2.2e-16	P value less than 5 percent hence significant evidence to reject a hypothesis that there is no intercept. Therefore, this represents baseline log wage when all predictors are at their reference levels.
Age	0.011	0.0001	< 2.2e-16	P value less than 5 percent hence significant evidence to reject a hypothesis that there is no impact by Age on wages. Thus this represents that for each additional year of age, the log weekly wage increases by 0.011 on average, holding all else constant.
EducCode	0.111	0.003	< 2.2e-16	P value less than 5 percent hence significant evidence to reject a hypothesis that there is no impact by EducCode on wages. Thus this represents that for each one-unit increase in education code, the log weekly wage increases by 0.111, holding all else constant.
LocationNotMetro	-0.087	0.018	2.03 e-06	P value less than 5 percent hence significant evidence to reject a hypothesis that there is no difference in impact between being employed in NotMet and Met areas on wages. Thus this represents that working in a nonmetropolitan area decreases the log weekly wage by 0.087 on average compared to a met area , holding all else constant.
RaceWhite	0.201	0.027	4.59e-14	P value less than 5 percent hence significant evidence to reject a hypothesis that there is no difference in impact between being white or black on wages. Thus this represents that white individuals have a log weekly wage that is 0.201 higher on average than black individuals, holding all else constant.
RegionNortheast	0.0904	0.021	1.53e-05	P value less than 5 percent hence significant evidence to reject a hypothesis that there is no difference in impact between employement in the Northeast and in the Midwest on wages. Thus this represent being employed in the Northeast increases the log weekly wage by 0.0904 compared to the Midwest.
RegionSouth	0.002	0.019	0.903	P value greater than 5 percent hence insufficient evidence to reject a hypothesis that there is no difference in impact between employment in the South and the Midwest on wages. Thus this represents that there is no significant difference in the log weekly wage between the South and the Midwest.
RegionWest	0.042	0.021	0.041	P value less than 5 percent hence significant evidence to reject a hypothesis that there is no difference in impact between employment in the West and the Midwest on wages. Thus this represents that being in the West increases the log weekly wage by 0.0423 compared to being employed in the Midwest, holding all else constant.

6.1 Evidence of Differences Between Black and White Workers

Coefficient for `RaceWhite`: $\beta_{\text{White}} = 0.201$

Interpretation: After accounting for all other variables, White workers earn approximately $100 \times (\exp(0.201) - 1) \approx 22.3\%$ more weekly than Black workers ($\beta = 0$ and hence $\exp \beta_{\text{black}} = 1$ so I can just look at the $\beta \pm 1.96 \times \text{SE}$ to get percentage confidence intervals).

Confidence Interval (95%): Using $SE = 0.027$:

$$CI = 0.201 \pm 1.96 \times 0.027 = (0.160, 0.289)$$

Conclusion: White workers earn between 16.0% and 28.9% more than Black workers, with strong evidence ($p < 2.2 \times 10^{-16}$).

6.2 Effect of Age

Coefficient for `Age`: $\beta_{\text{Age}} = 0.011$

Interpretation: For each additional year of age, weekly wages increase by approximately $100 \times (\exp(0.011) - 1) \approx 1.11\%$, holding all else constant.

Confidence Interval (95%): Using $SE = 0.001$:

$$CI = 0.011 \pm 1.96 \times 0.001 = (0.0090, 0.0130)$$

Conclusion: Age positively impacts wages, with a modest (improvement by 0.9% to 1.3% with 95% certainty) but statistically significant effect ($p < 2.2 \times 10^{-16}$).

6.3 Effect of Education Code

Coefficient for `EducCode`: $\beta_{\text{EducCode}} = 0.111$

Interpretation: Each additional education level increases weekly wages by $100 \times (\exp(0.111) - 1) \approx 11.7\%$, holding other variables constant.

Confidence Interval (95%): Using $SE = 0.003$:

$$CI = 0.111 \pm 1.96 \times 0.003 = (0.105, 0.117)$$

Conclusion: Education significantly boosts wages(by 10.5% to 11.7% with 95% certainty), with consistent gains across levels with strong evidence ($p < 2.2 \times 10^{-16}$)

6.4 Effect of Location

Coefficient for `LocationNotMetro`: $\beta_{\text{NotMetro}} = -0.087$

Interpretation: Living in a non-metropolitan area decreases weekly wages by $100 \times (\exp(-0.087) - 1) \approx -8.33\%$, holding other factors constant.

Confidence Interval (95%): Using $SE = 0.018$:

$$CI = -0.087 \pm 1.96 \times 0.018 = (-0.122, -0.052)$$

Conclusion: Workers in non-metro areas earn significantly less (between 5.2% to 12.2% less with 95% certainty) than their metro counterparts with strong evidence ($p < 2.03 \times 10^{-6}$)

6.5 Regional Effects

RegionMidwest (Baseline region):

- **RegionNortheast:** $\beta = 0.0904$ and SE = 0.021: Wages are +9.46% higher than in the Midwest. $CI : (4.92\%, 13.2\%)$. Therefore, workers in Northeast region earn considerably more (between 4.92% and 13.2% with 95% certainty) than their Midwest counterparts with strong evidence ($p < 1.52 \times 10^{-5}$)
- **RegionSouth:** $\beta = 0.002$ and SE = 0.019: No statistically significant difference in wages compared to the Midwest ($p = 0.9031$).
- **RegionWest:** $\beta = 0.042$ and SE = 0.021: Wages are +4.29% higher than in the Midwest. $CI : (0.084\%, 8.32\%)$. Therefore, workers in the West region earn moderately more (between 0.084% and 8.32% with 95% certainty) than their Midwest counterparts with just about sufficient evidence ($p = 0.041$)

6.6 Summary Table

Table 2: Summary of Effects on Weekly Wages

Variable	Effect (%)	95% CI (%)	Interpretation
RaceWhite	+22.3	(16.0, 28.9)	White workers earn significantly more than Black workers.
Age	+1.11	(0.90, 1.30)	Each additional year of age increases wages slightly.
EducCode	+11.1	(10.5, 11.7)	Higher education levels significantly increase wages.
LocationNotMetro	-8.33	(-12.2, -5.2)	Workers working in non-metropolitan areas earn significantly less than those in metropolitan areas.
RegionNortheast	+9.46	(4.92, 13.2)	Wages in the Northeast are substantially higher than in the Midwest.
RegionSouth	No effect	Not significant	Wages in the South do not significantly differ from the Midwest.
RegionWest	+4.29	(0.084, 8.32)	Wages in the West are moderately higher than in the Midwest.

6.7 Conclusions relevant to the question posed

- **Race:** White workers earn approximately +22.2% more than Black workers on average, a difference that remains after adjusting for all other variables.
- **Education:** Each additional education level boosts wages by +11.77%, highlighting the strong economic returns of education.
- **Location:** Living in non-metropolitan areas reduces wages significantly (−8.34%).
- **Regional Effects:** The Northeast and Midwest offer the highest wages (Northeast highest), while the South shows no significant difference, and the West has modestly higher wages than the Midwest.
- **Age:** whilst the effect in minor, there is still a +1.11% increase in wages per additional year of age increase.

Appendix

```
wages <- read.csv("https://www.stats.ox.ac.uk/~laws/SB1/data/wages.csv",
    stringsAsFactors = TRUE )
library(ggplot2)
library(dplyr)
# Explanatory data analysis
str(wages)
summary(wages)
##PLOTS NEEDED:
# Helper function to create individual boxplots
create_boxplot <- function(group_var, title) {
  means <- wages %>%
    group_by(.data[[group_var]]) %>%
    summarise(MeanWage = mean(Wage, na.rm = TRUE))
  ggplot(wages, aes(x = .data[[group_var]], y = Wage, fill = .data[[group_var]])) +
    geom_boxplot(outlier.shape = NA, alpha = 0.7) +
```

```
    stat_summary(fun = "mean", geom = "point", shape = 23, size = 3, fill = "red")
        +
    geom_text(data = means, aes(x = .data[[group_var]], y = MeanWage, label = round
        (MeanWage, 1)),
              vjust = -0.5, color = "blue", size = 3) +
    geom_hline(yintercept = 1056.4, color = "red", linetype = "dashed", size = 1)+
    labs(
      title = title,
      x = group_var,
      y = "Weekly Earnings"
    ) +
    theme_minimal() +
    theme(
      plot.title = element_text(hjust = 0.5, size = 12),
      axis.title = element_text(size = 10),
      legend.position = "none"
    )
}
#Create and display individual plots
plot_race <- create_boxplot("Race", "Weekly Earnings by Race")
plot_location <- create_boxplot("Location", "Weekly Earnings by Location")
plot_educ <- create_boxplot("EducCateg", "Weekly Earnings by Education Category") +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0.5)
  )
plot_region <- create_boxplot("Region", "Weekly Earnings by Region")
print(plot_race)
print(plot_location)
print(plot_educ)
print(plot_region)
ggplot(wages, aes(x = factor(EducCode), y = Wage)) +
  geom_boxplot(outlier.shape = NA, fill = "skyblue", alpha = 0.7) +
  stat_summary(fun = "mean", geom = "point", shape = 23, size = 3, fill = "red") +
  geom_text(data = educ_means, aes(x = factor(EducCode), y = MeanWage, label =
      round(MeanWage, 1)),
            vjust = -0.5, color = "blue", size = 3) +
  geom_hline(yintercept = 1056.4, color = "red", linetype = "dashed", size = 1) +
  labs(
    title = "Weekly Earnings vs Education codes",
    x = "Education Code",
    y = "Weekly Earnings"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1))
#Scatter plot og average wage for each age against each age
average_wage_by_age <- aggregate(Wage ~ Age, wages, mean)
ggplot(average_wage_by_age, aes(x = Age, y = Wage)) +
  geom_line(color = "blue") +
  geom_point(color = "red") +
  geom_hline(yintercept = 1056.4, color = "red", linetype = "dashed", size = 1) +
  labs(title = "Average Wage by Age",
       x = "Age",
       y = "Average Wage") +
  theme_minimal()
# Function to calculate quartiles and IQR for a specified group
calculate_quartiles_iqr <- function(group_var) {
  wages %>%
    group_by(.data[[group_var]]) %>%
    summarise(
      Q1 = quantile(Wage, 0.25, na.rm = TRUE),
      Median = median(Wage, na.rm = TRUE),
      Q3 = quantile(Wage, 0.75, na.rm = TRUE),
```

```
      IQR = IQR(Wage, na.rm = TRUE),
      .groups = 'drop'
    ) %>%
    arrange(.data[[group_var]])
}
#Calculate for "Race"
quartiles_race <- calculate_quartiles_iqr("Race")
#Calculate for "Location"
quartiles_location <- calculate_quartiles_iqr("Location")
quartiles_educ <- calculate_quartiles_iqr("EducCateg")
#Calculate for "Region"
quartiles_region <- calculate_quartiles_iqr("Region")
# Display results
print(quartiles_race)
print(quartiles_location)
print(quartiles_educ)
print(quartiles_region)

########
##NORMAL LINEAR MODEL
#Initial model
lm1 <- lm(Wage ~ Age + EducCode + Location + Region + Race, data = wages)
summary(lm1)
# Residual plots
par(mfrow = c(2, 2))
plot(initial_model)
# Calculate studentised residuals
studentised_residuals <- rstudent(initial_model)
# Set up a 2x2 plotting grid
par(mfrow = c(1, 2))
# 1. Studentised Residuals vs Fitted Values
plot(fitted(initial_model), studentised_residuals,
     xlab = "Fitted Values",
     ylab = "Studentised Residuals",
     main = "Studentised Residuals vs Fitted Values")
abline(h = 0, col = "red", lty = 2)  # Add horizontal reference line at 0
# 2. Normal Q-Q Plot
qqnorm(studentised_residuals,
       main = "Normal Q-Q Plot of Studentised Residuals")
qqline(studentised_residuals, col = "red")  # Add a reference line
# Box-Cox transformation to check if a transformation of Wage improves model fit
par(mfrow = c(1, 1))
boxcox_result <- boxcox(initial_model, lambda = seq(-2, 2, by = 0.1))
best_lambda <- boxcox_result$x[which.max(boxcox_result$y)]
cat("Best lambda for Box-Cox transformation:", best_lambda, "\n")
# Apply the logarithmic transformation for Wage
wages$LogWage <- log(wages$Wage)
# Fit the new linear model with the transformed response variable
lm2 <- lm(LogWage ~ Age + EducCode + Location + Race + Region, data = wages)
# Summary of the new model
summary(lm2)
# Calculate studentised residuals
studentised_residuals <- rstudent(lm2)
# Set up a 1x2 plotting grid
par(mfrow = c(1, 2))  # Arrange plots in a 2x2 layout
# 1. Studentised Residuals vs Fitted Values
plot(fitted(lm2), studentised_residuals,
     xlab = "Fitted Values",
     ylab = "Studentised Residuals",
     main = "Studentised Residuals vs Fitted Values")
abline(h = 0, col = "red", lty = 2)
abline(h = c(-3, 3), col = "red", lty = 2)# Add a horizontal reference line at 0
```

```r
# 2. Normal Q-Q Plot of Studentised Residuals
qqnorm(studentised_residuals,
       main = "Normal Q-Q Plot of Studentised Residuals")
qqline(studentised_residuals, col = "red")  # Add a reference line
# Number of observations (n) and predictors (p)
n <- nrow(wages)
p <- length(coef(lm2))
# Calculate Cook's Distance
cook_dist <- cooks.distance(lm2)
# Threshold for Cook's Distance
cook_threshold <- 8 / (n - 2 * p)
# Identify observations exceeding the Cook's Distance threshold
influential_obs <- which(cook_dist > cook_threshold)
# Cook's Distance Plot
par(mfrow = c(1, 1))
plot(cook_dist,
     type = "h",
     main = "Cook's Distance with Threshold",
     xlab = "Observation",
     ylab = "Cook's Distance",
     ylim = c(0, max(c(cook_dist, cook_threshold))))  # Ensure the threshold line
         fits in the plot
abline(h = cook_threshold, col = "red", lty = 2, lwd = 2)  # Add threshold line
# Print observations being removed
cat("Removing influential observations:\n")
print(influential_obs)
# Remove influential observations
wages_clean <- wages[-influential_obs, ]
# Refit the model without influential observations
lm2_clean <- lm(LogWage ~ Age + EducCode + Location + Race + Region, data = wages_
    clean)
# Summary of the new model
summary(lm2_clean)
# Generate Studentised Residuals Plots for the Updated Model
studentised_residuals <- rstudent(lm2_clean)
# Set up a 2x2 plotting grid
par(mfrow = c(1, 2))
# 1. Studentised Residuals vs Fitted Values
plot(fitted(lm2_clean), studentised_residuals,
     xlab = "Fitted Values",
     ylab = "Studentised Residuals",
     main = "Studentised Residuals vs Fitted Values")
abline(h = 0, col = "red", lty = 2)
abline(h = c(-3, 3), col = "red", lty = 2)# Add a horizontal reference line at 0
# 2. Normal Q-Q Plot of Studentised Residuals
qqnorm(studentised_residuals,
       main = "Normal Q-Q Plot of Studentised Residuals")
qqline(studentised_residuals, col = "red")

########
##CODE VS CATEGORY
# Model with Education as Numerical
lm_edu_num <- lm(LogWage ~ Age + EducCode + Location + Race + Region, data = wages)
summary(lm_edu_num)
# Model with Education as Categorical
lm_edu_cat <- lm(LogWage ~ Age + EducCateg + Location + Race + Region, data = wages
    )
summary(lm_edu_cat)
# Compare AIC and BIC
cat("BIC for numerical model:", BIC(lm_edu_num), "\n")
cat("BIC for categorical model:", BIC(lm_edu_cat), "\n")
```